

Secret Communication with Plausible Deniability

Stony Brook International Conference

Xiaoyu Cheng

Florida State University

Yonggyun (YG) Kim

Florida State University

Pak Hang (Michael) Tam

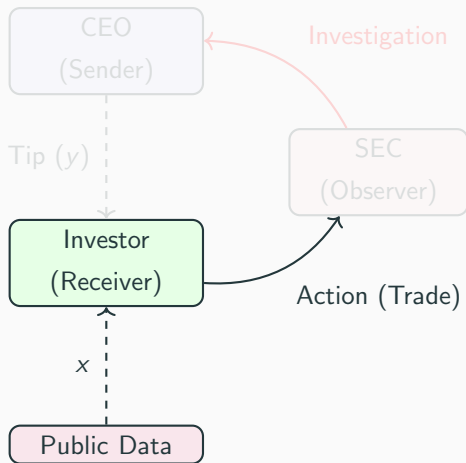
University of North Carolina, Chapel Hill

July 22, 2026

Motivation

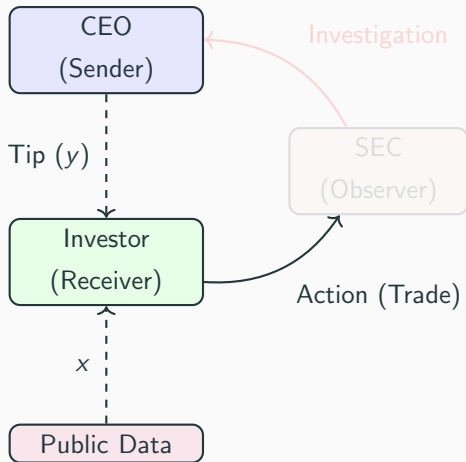
Motivation: The Cautious Insider

- **Investor** trades a stock based on public baseline data (e.g., earnings reports).
- **CEO** wants to secretly pass a valuable tip to this favored investor.
 - CEO is cautious about SEC scrutiny.
- **Friction:** CEO *cannot control* the investor's actions.
 - With full information, the investor may trade aggressively against public data, triggering an SEC investigation.
 - The CEO must design the tip so the investor's optimal response remains safe.



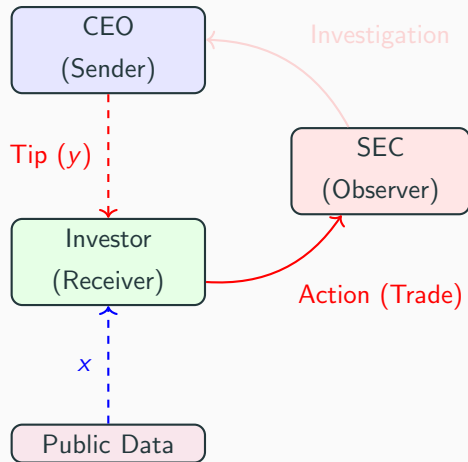
Motivation: The Cautious Insider

- **Investor** trades a stock based on public baseline data (e.g., earnings reports).
- **CEO** wants to secretly pass a valuable tip to this favored investor.
 - CEO is cautious about SEC scrutiny.
- **Friction:** CEO *cannot control* the investor's actions.
 - With full information, the investor may trade aggressively against public data, triggering an SEC investigation.
 - The CEO must design the tip so the investor's optimal response remains safe.



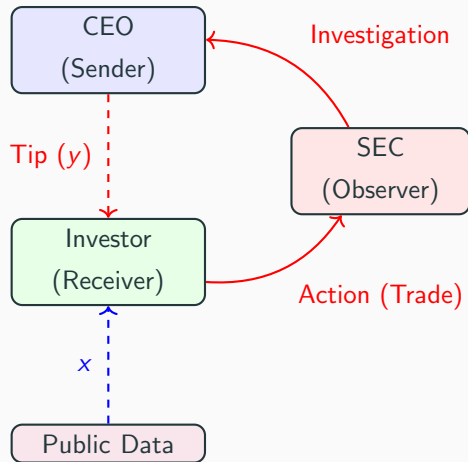
Motivation: The Cautious Insider

- **Investor** trades a stock based on public baseline data (e.g., earnings reports).
- **CEO** wants to secretly pass a valuable tip to this favored investor.
 - CEO is cautious about SEC scrutiny.
- **Friction:** CEO *cannot control* the investor's actions.
 - With full information, the investor may trade aggressively against public data, triggering an SEC investigation.
 - The CEO must design the tip so the investor's optimal response remains safe.



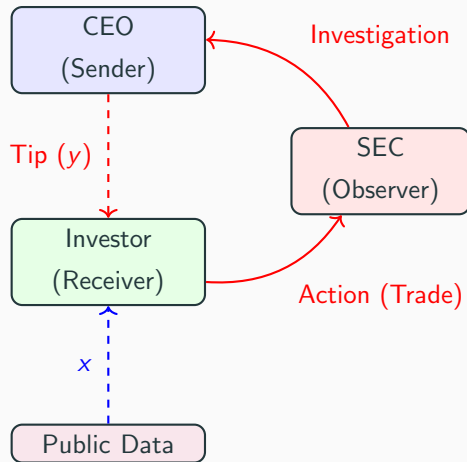
Motivation: The Cautious Insider

- **Investor** trades a stock based on public baseline data (e.g., earnings reports).
- **CEO** wants to secretly pass a valuable tip to this favored investor.
 - CEO is cautious about SEC scrutiny.
- **Friction:** CEO *cannot control* the investor's actions.
 - With full information, the investor may trade aggressively against public data, triggering an SEC investigation.
 - The CEO must design the tip so the investor's optimal response remains safe.



Motivation: The Cautious Insider

- **Investor** trades a stock based on public baseline data (e.g., earnings reports).
- **CEO** wants to secretly pass a valuable tip to this favored investor.
 - CEO is cautious about SEC scrutiny.
- **Friction:** CEO *cannot control* the investor's actions.
 - With full information, the investor may trade aggressively against public data, triggering an SEC investigation.
 - The CEO must design the tip so the investor's optimal response remains safe.



Motivation: What Makes Communication “Safe”?

- Sender designs information under following constraints:
 1. **Secrecy**: Message must be statistically independent of state. [Shannon, 1949]
 - The tip itself cannot be incriminating evidence if intercepted.
 2. **Plausible Deniability (PD)**: Receiver's induced action must be justifiable using *only* the baseline public data.
 - Even if the SEC investigates, the CEO and the investor can credibly claim:
“Given the public earnings report, the investor's trade makes sense.”
- Other Applications: whistleblowing to regulators, covert intelligence in military operations, and diplomatic backchannels.
- **Question**: What is the *maximally informative* joint information structure the Sender can design under these dual constraints?

Motivation: What Makes Communication “Safe”?

- Sender designs information under following constraints:
 1. **Secrecy**: Message must be statistically independent of state. [Shannon, 1949]
 - The tip itself cannot be incriminating evidence if intercepted.
 2. **Plausible Deniability (PD)**: Receiver's induced action must be justifiable using *only* the baseline public data.
 - Even if the SEC investigates, the CEO and the investor can credibly claim:
“Given the public earnings report, the investor's trade makes sense.”
- Other Applications: whistleblowing to regulators, covert intelligence in military operations, and diplomatic backchannels.
- **Question**: What is the *maximally informative* joint information structure the Sender can design under these dual constraints?

Motivation: What Makes Communication “Safe”?

- Sender designs information under following constraints:
 1. **Secrecy**: Message must be statistically independent of state. [Shannon, 1949]
 - The tip itself cannot be incriminating evidence if intercepted.
 2. **Plausible Deniability (PD)**: Receiver's induced action must be justifiable using *only* the baseline public data.
 - Even if the SEC investigates, the CEO and the investor can credibly claim: “Given the public earnings report, the investor's trade makes sense.”
- Other Applications: whistleblowing to regulators, covert intelligence in military operations, and diplomatic backchannels.
- **Question**: What is the *maximally informative* joint information structure the Sender can design under these dual constraints?

Motivation: What Makes Communication “Safe”?

- Sender designs information under following constraints:
 1. **Secrecy**: Message must be statistically independent of state. [Shannon, 1949]
 - The tip itself cannot be incriminating evidence if intercepted.
 2. **Plausible Deniability (PD)**: Receiver's induced action must be justifiable using *only* the baseline public data.
 - Even if the SEC investigates, the CEO and the investor can credibly claim:
“Given the public earnings report, the investor's trade makes sense.”
- Other Applications: whistleblowing to regulators, covert intelligence in military operations, and diplomatic backchannels.
- **Question**: What is the *maximally informative* joint information structure the Sender can design under these dual constraints?

Motivation: What Makes Communication “Safe”?

- Sender designs information under following constraints:
 1. **Secrecy**: Message must be statistically independent of state. [Shannon, 1949]
 - The tip itself cannot be incriminating evidence if intercepted.
 2. **Plausible Deniability (PD)**: Receiver's induced action must be justifiable using *only* the baseline public data.
 - Even if the SEC investigates, the CEO and the investor can credibly claim:
“Given the public earnings report, the investor's trade makes sense.”
- Other Applications: whistleblowing to regulators, covert intelligence in military operations, and diplomatic backchannels.
- **Question**: What is the *maximally informative* joint information structure the Sender can design under these dual constraints?

Model Setup

Setup: Preliminaries

- **Players:** Sender, Receiver and (outside) Observers
 - Observers do not play any action; they simply impose the constraints
- **State and Prior:** State $\omega \in \Omega := \{\omega_1, \dots, \omega_N\}$ with full-support prior $\mu \in \Delta(\Omega)$
 - States are ordered: $\omega_1 < \dots < \omega_N$
- **Baseline Information (Public):** Receiver observes a baseline message $x \in X$ generated by an exogenous information structure $f : \Omega \rightarrow \Delta(X)$.
- **Sender's Design (Shared):** Sender designs a joint information structure $h : \Omega \rightarrow \Delta(X \times Y)$, generating an additional message $y \in Y$.
 - The X -marginal must equal f : $\sum_y h(x, y|\omega) = f(x|\omega)$.
 - The Y -marginal is publicly observable: $g(y|\omega) = \sum_x h(x, y|\omega)$.
 - The choice of h is shared between Sender and Receiver, but hidden from Observer.

Setup: Preliminaries

- **Players:** Sender, Receiver and (outside) Observers
 - Observers do not play any action; they simply impose the constraints
- **State and Prior:** State $\omega \in \Omega := \{\omega_1, \dots, \omega_N\}$ with full-support prior $\mu \in \Delta(\Omega)$
 - States are ordered: $\omega_1 < \dots < \omega_N$
- **Baseline Information (Public):** Receiver observes a baseline message $x \in X$ generated by an exogenous information structure $f : \Omega \rightarrow \Delta(X)$.
- **Sender's Design (Shared):** Sender designs a joint information structure $h : \Omega \rightarrow \Delta(X \times Y)$, generating an additional message $y \in Y$.
 - The X -marginal must equal f : $\sum_y h(x, y|\omega) = f(x|\omega)$.
 - The Y -marginal is publicly observable: $g(y|\omega) = \sum_x h(x, y|\omega)$.
 - The choice of h is shared between Sender and Receiver, but hidden from Observer.

Setup: Preliminaries

- **Players:** Sender, Receiver and (outside) Observers
 - Observers do not play any action; they simply impose the constraints
- **State and Prior:** State $\omega \in \Omega := \{\omega_1, \dots, \omega_N\}$ with full-support prior $\mu \in \Delta(\Omega)$
 - States are ordered: $\omega_1 < \dots < \omega_N$
- **Baseline Information (Public):** Receiver observes a baseline message $x \in X$ generated by an exogenous information structure $f : \Omega \rightarrow \Delta(X)$.
- **Sender's Design (Shared):** Sender designs a joint information structure $h : \Omega \rightarrow \Delta(X \times Y)$, generating an additional message $y \in Y$.
 - The X -marginal must equal f : $\sum_y h(x, y|\omega) = f(x|\omega)$.
 - The Y -marginal is publicly observable: $g(y|\omega) = \sum_x h(x, y|\omega)$.
 - The choice of h is shared between Sender and Receiver, but hidden from Observer.

Setup: Preliminaries

- **Players:** Sender, Receiver and (outside) Observers
 - Observers do not play any action; they simply impose the constraints
- **State and Prior:** State $\omega \in \Omega := \{\omega_1, \dots, \omega_N\}$ with full-support prior $\mu \in \Delta(\Omega)$
 - States are ordered: $\omega_1 < \dots < \omega_N$
- **Baseline Information (Public):** Receiver observes a baseline message $x \in X$ generated by an exogenous information structure $f : \Omega \rightarrow \Delta(X)$.
- **Sender's Design (Shared):** Sender designs a joint information structure $h : \Omega \rightarrow \Delta(X \times Y)$, generating an additional message $y \in Y$.
 - The X -marginal must equal f : $\sum_y h(x, y|\omega) = f(x|\omega)$.
 - The Y -marginal is publicly observable: $g(y|\omega) = \sum_x h(x, y|\omega)$.
 - The choice of h is shared between Sender and Receiver, but hidden from Observer.

Setup: Preliminaries

- **Players:** Sender, Receiver and (outside) Observers
 - Observers do not play any action; they simply impose the constraints
- **State and Prior:** State $\omega \in \Omega := \{\omega_1, \dots, \omega_N\}$ with full-support prior $\mu \in \Delta(\Omega)$
 - States are ordered: $\omega_1 < \dots < \omega_N$
- **Baseline Information (Public):** Receiver observes a baseline message $x \in X$ generated by an exogenous information structure $f : \Omega \rightarrow \Delta(X)$.
- **Sender's Design (Shared):** Sender designs a joint information structure $h : \Omega \rightarrow \Delta(X \times Y)$, generating an additional message $y \in Y$.
 - The X -marginal must equal f : $\sum_y h(x, y|\omega) = f(x|\omega)$.
 - The Y -marginal is publicly observable: $g(y|\omega) = \sum_x h(x, y|\omega)$.
 - The choice of h is shared between Sender and Receiver, but hidden from Observer.

Setup: Preferences

- The set of actions (by Receiver) is an ordered set $A = \{a_{-I}, \dots, a_0, \dots, a_L\}$.
 - a_0 is the default action—unique optimal under the prior μ .
- Receiver's utility function $u : A \times \Omega \rightarrow \mathbb{R}$ satisfies the *single-crossing* property (SCP) if for any actions $a' > a$ and states $\omega' > \omega$:

$$u(a', \omega) \geq (>) u(a, \omega) \quad \implies \quad u(a', \omega') \geq (>) u(a, \omega').$$

- The optimal action is increasing in states
- Let $\mathcal{U}$ denote the class of utilities satisfying above two properties
 - $u \in \mathcal{U}$ is public knowledge, but exact form of u is unknown to Sender and Observer

Setup: Preferences

- The set of actions (by Receiver) is an ordered set $A = \{a_{-I}, \dots, a_0, \dots, a_L\}$.
 - a_0 is the default action—unique optimal under the prior μ .
- Receiver's utility function $u : A \times \Omega \rightarrow \mathbb{R}$ satisfies the *single-crossing* property (SCP) if for any actions $a' > a$ and states $\omega' > \omega$:

$$u(a', \omega) \geq (>) u(a, \omega) \quad \implies \quad u(a', \omega') \geq (>) u(a, \omega').$$

- The optimal action is increasing in states
- Let $\mathcal{U}$ denote the class of utilities satisfying above two properties
 - $u \in \mathcal{U}$ is public knowledge, but exact form of u is unknown to Sender and Observer

Setup: Preferences

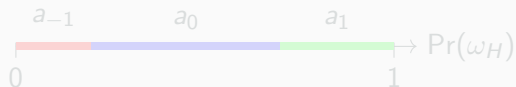
- The set of actions (by Receiver) is an ordered set $A = \{a_{-I}, \dots, a_0, \dots, a_L\}$.
 - a_0 is the default action—unique optimal under the prior μ .
- Receiver's utility function $u : A \times \Omega \rightarrow \mathbb{R}$ satisfies the *single-crossing* property (SCP) if for any actions $a' > a$ and states $\omega' > \omega$:

$$u(a', \omega) \geq (>) u(a, \omega) \quad \implies \quad u(a', \omega') \geq (>) u(a, \omega').$$

- The optimal action is increasing in states
- Let $\mathcal{U}$ denote the class of utilities satisfying above two properties
 - $u \in \mathcal{U}$ is public knowledge, but exact form of u is unknown to Sender and Observer

Setup: Optimal Actions

- Optimal actions given $u \in \mathcal{U}$:
 - $a_{x,y}(u)$ given (x, y)
 - $a_x(u)$ given x
 - $a_\emptyset(u) = a_0$ is the default action
- Example: $\Omega = \{\omega_L, \omega_H\}$, $A = \{a_{-1}, a_0, a_1\}$



- $a_\emptyset(u) = a_0$
- $a_x(u) = a_1$
- $a_{x,y}(u) = a_{-1}$

Setup: Optimal Actions

- Optimal actions given $u \in \mathcal{U}$:
 - $a_{x,y}(u)$ given (x, y)
 - $a_x(u)$ given x
 - $a_\emptyset(u) = a_0$ is the default action
- Example: $\Omega = \{\omega_L, \omega_H\}$, $A = \{a_{-1}, a_0, a_1\}$



- $a_\emptyset(u) = a_0$
- $a_x(u) = a_1$
- $a_{x,y}(u) = a_{-1}$

Setup: Optimal Actions

- Optimal actions given $u \in \mathcal{U}$:
 - $a_{x,y}(u)$ given (x, y)
 - $a_x(u)$ given x
 - $a_\emptyset(u) = a_0$ is the default action
- Example: $\Omega = \{\omega_L, \omega_H\}$, $A = \{a_{-1}, a_0, a_1\}$



- $a_\emptyset(u) = a_0$
- $a_x(u) = a_1$
- $a_{x,y}(u) = a_{-1}$

Setup: Optimal Actions

- Optimal actions given $u \in \mathcal{U}$:
 - $a_{x,y}(u)$ given (x, y)
 - $a_x(u)$ given x
 - $a_\emptyset(u) = a_0$ is the default action
- Example: $\Omega = \{\omega_L, \omega_H\}$, $A = \{a_{-1}, a_0, a_1\}$



- $a_\emptyset(u) = a_0$
- $a_x(u) = a_1$
- $a_{x,y}(u) = a_{-1}$

Setup: Optimal Actions

- Optimal actions given $u \in \mathcal{U}$:
 - $a_{x,y}(u)$ given (x, y)
 - $a_x(u)$ given x
 - $a_\emptyset(u) = a_0$ is the default action
- Example: $\Omega = \{\omega_L, \omega_H\}$, $A = \{a_{-1}, a_0, a_1\}$



- $a_\emptyset(u) = a_0$
- $a_x(u) = a_1$
- $a_{x,y}(u) = a_{-1}$

Setup: Designing Information

The sender's objective is to design a joint information as informative as possible subject to constraints

1. $h : \Omega \rightarrow \Delta(X \times Y)$ satisfies **secrecy** iff $g : \Omega \rightarrow \Delta(Y)$ is uninformative
2. $h : \Omega \rightarrow \Delta(X \times Y)$ satisfies **plausible deniability** iff for all $u \in \mathcal{U}$ and $(x, y) \in X \times Y$, there exists $\tilde{u} \in \mathcal{U}$ such that

$$a_{x,y}(u) = a_x(\tilde{u}).$$

- Violation of Plausible Deniability:



Setup: Designing Information

The sender's objective is to design a joint information as informative as possible subject to constraints

1. $h : \Omega \rightarrow \Delta(X \times Y)$ satisfies **secrecy** iff $g : \Omega \rightarrow \Delta(Y)$ is uninformative
2. $h : \Omega \rightarrow \Delta(X \times Y)$ satisfies **plausible deniability** iff for all $u \in \mathcal{U}$ and $(x, y) \in X \times Y$, there exists $\tilde{u} \in \mathcal{U}$ such that

$$a_{x,y}(u) = a_x(\tilde{u}).$$

- Violation of Plausible Deniability:



Setup: Designing Information

The sender's objective is to design a joint information as informative as possible subject to constraints

1. $h : \Omega \rightarrow \Delta(X \times Y)$ satisfies **secrecy** iff $g : \Omega \rightarrow \Delta(Y)$ is uninformative
2. $h : \Omega \rightarrow \Delta(X \times Y)$ satisfies **plausible deniability** iff for all $u \in \mathcal{U}$ and $(x, y) \in X \times Y$, there exists $\tilde{u} \in \mathcal{U}$ such that

$$a_{x,y}(u) = a_x(\tilde{u}).$$

- Violation of Plausible Deniability:



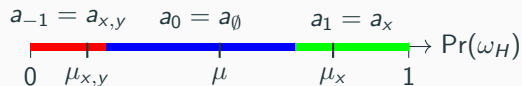
Setup: Designing Information

The sender's objective is to design a joint information as informative as possible subject to constraints

1. $h : \Omega \rightarrow \Delta(X \times Y)$ satisfies **secrecy** iff $g : \Omega \rightarrow \Delta(Y)$ is uninformative
2. $h : \Omega \rightarrow \Delta(X \times Y)$ satisfies **plausible deniability** iff for all $u \in \mathcal{U}$ and $(x, y) \in X \times Y$, there exists $\tilde{u} \in \mathcal{U}$ such that

$$a_{x,y}(u) = a_x(\tilde{u}).$$

- Violation of Plausible Deniability:



Setup: Greatest Element

- $\mathcal{H}^{SPD}$: the set of joint information structures satisfying both secrecy (**S**) and plausible deniability (**PD**). Similar for $\mathcal{H}^{PD}$ and $\mathcal{H}^S$.
- **Dominance**: we say h dominates h' if for all $u \in \mathcal{U}$, Receiver's expected utility is weakly greater under h than under h' .
- h is the **SPD-greatest element** in $\mathcal{H}^{SPD}$ if h dominates h' for all $h' \in \mathcal{H}^{SPD}$. Similar for **S-greatest** and **PD-greatest**.
- The goal is to find the SPD-greatest element.

Setup: Greatest Element

- $\mathcal{H}^{SPD}$: the set of joint information structures satisfying both secrecy (**S**) and plausible deniability (**PD**). Similar for $\mathcal{H}^{PD}$ and $\mathcal{H}^S$.
- **Dominance**: we say h dominates h' if for all $u \in \mathcal{U}$, Receiver's expected utility is weakly greater under h than under h' .
- h is the **SPD-greatest element** in $\mathcal{H}^{SPD}$ if h dominates h' for all $h' \in \mathcal{H}^{SPD}$. Similar for **S-greatest** and **PD-greatest**.
- The goal is to find the SPD-greatest element.

Setup: Greatest Element

- $\mathcal{H}^{SPD}$: the set of joint information structures satisfying both secrecy (**S**) and plausible deniability (**PD**). Similar for $\mathcal{H}^{PD}$ and $\mathcal{H}^S$.
- **Dominance**: we say h dominates h' if for all $u \in \mathcal{U}$, Receiver's expected utility is weakly greater under h than under h' .
- h is the **SPD-greatest element** in $\mathcal{H}^{SPD}$ if h dominates h' for all $h' \in \mathcal{H}^{SPD}$. Similar for **S-greatest** and **PD-greatest**.
- The goal is to find the SPD-greatest element.

Setup: Greatest Element

- $\mathcal{H}^{SPD}$: the set of joint information structures satisfying both secrecy (**S**) and plausible deniability (**PD**). Similar for $\mathcal{H}^{PD}$ and $\mathcal{H}^S$.
- **Dominance**: we say h dominates h' if for all $u \in \mathcal{U}$, Receiver's expected utility is weakly greater under h than under h' .
- h is the **SPD-greatest element** in $\mathcal{H}^{SPD}$ if h dominates h' for all $h' \in \mathcal{H}^{SPD}$. Similar for **S-greatest** and **PD-greatest**.
- The goal is to find the SPD-greatest element.

2 by 2 Example

2 by 2 Example: Optimal Secret Information (w/o PD)

- $\Omega = \{\omega_L, \omega_H\}$, $X = \{L, H\}$

	L	H
ω_L	.8	.2
ω_H	.2	.8

2 by 2 Example: Optimal Secret Information (w/o PD)

- $\Omega = \{\omega_L, \omega_H\}$, $X = \{L, H\}$, $Y = \{T(ucan), F(alcon)\}$

	LT	HF	LF	HT
ω_L	.8	.2	0	0
ω_H	0	0	.2	.8

2 by 2 Example: Optimal Secret Information (w/o PD)

- $\Omega = \{\omega_L, \omega_H\}$, $X = \{L, H\}$, $Y = \{T(ucan), F(alcon)\}$

	LT	HF	LF	HT		T	F
ω_L	.8	.2	0	0	ω_L	.8	.2
ω_H	0	0	.2	.8	ω_H	.8	.2

- This joint information structure is most informative and satisfies *secrecy* (S-greatest element), but it violates the *plausible deniability*



2 by 2 Example: Optimal Secret Information (w/o PD)

- $\Omega = \{\omega_L, \omega_H\}$, $X = \{L, H\}$, $Y = \{T(ucan), F(alcon)\}$

	LT	HF	LF	HT		T	F
ω_L	.8	.2	0	0	ω_L	.8	.2
ω_H	0	0	.2	.8	ω_H	.8	.2

- This joint information structure is most informative and satisfies *secrecy* (S-greatest element), but it violates the *plausible deniability*



2 by 2 Example: S-greatest element

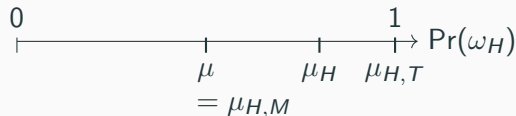
- $\Omega = \{\omega_L, \omega_H\}$, $X = \{L, H\}$, $Y = \{T, M(anatee, eaningless)\}$

	LT	HM	LM	HT
ω_L	.6	.2	.2	0
ω_H	0	.2	.2	.6

2 by 2 Example: S-greatest element

- $\Omega = \{\omega_L, \omega_H\}$, $X = \{L, H\}$, $Y = \{T, M(\text{anatee}, \text{eaningless})\}$

	LT	HM	LM	HT		T	M
ω_L	.6	.2	.2	0	ω_L	.6	.4
ω_H	0	.2	.2	.6	ω_H	.6	.4



- This joint information structure satisfies secrecy and plausible deniability,

2 by 2 Example: S-greatest element

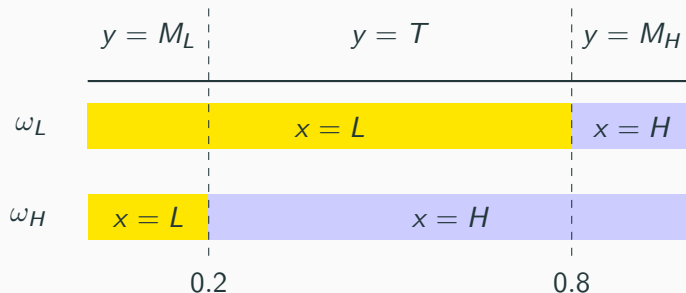
- $\Omega = \{\omega_L, \omega_H\}$, $X = \{L, H\}$, $Y = \{T, M(\text{anatee}, \text{eaningless})\}$

	LT	HM	LM	HT		T	M
ω_L	.6	.2	.2	0	ω_L	.6	.4
ω_H	0	.2	.2	.6	ω_H	.6	.4

- This joint information structure satisfies secrecy and plausible deniability, and it is the **SPD-greatest element**.

2 by 2 Example: Secrecy and Signal Representation

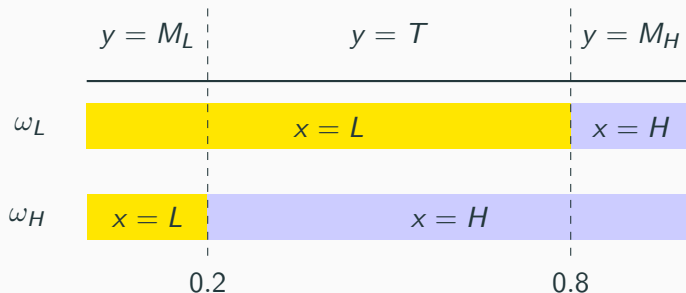
- To implement secrecy, it is useful to represent information as a signal
 $\psi : \Omega \times [0, 1] \rightarrow X$ (a partition of $\Omega \times [0, 1]$) [Green, Stokey (2022), Strack, Yang (2024)].



- Key observation:** The **ordered**-signal induces the SPD-greatest information in the 2 by 2 case

2 by 2 Example: Secrecy and Signal Representation

- To implement secrecy, it is useful to represent information as a signal
 $\psi : \Omega \times [0, 1] \rightarrow X$ (a partition of $\Omega \times [0, 1]$) [Green, Stokey (2022), Strack, Yang (2024)].



- Key observation:** The **ordered**-signal induces the SPD-greatest information in the 2 by 2 case

General Setup

Rationalizability and Plausible Deniability

Given $\phi : \Omega \rightarrow \Delta(Z)$ and $z \in \text{supp}(\phi)$,

- an action $a \in A$ is **rationalizable** at z (under ϕ) if there exists $u \in \mathcal{U}$ such that a is optimal under the posterior induced by observing z ;
- $\mathcal{R}_\phi(z) \subseteq A$ denote the set of rationalizable actions at z

Lemma 1

h satisfies plausible deniability if and only if for all $(x, y) \in \text{supp}(h)$,

$$\mathcal{R}_h(x, y) \subseteq \mathcal{R}_f(x).$$

Rationalizability and Plausible Deniability

Given $\phi : \Omega \rightarrow \Delta(Z)$ and $z \in \text{supp}(\phi)$,

- an action $a \in A$ is **rationalizable** at z (under ϕ) if there exists $u \in \mathcal{U}$ such that a is optimal under the posterior induced by observing z ;
- $\mathcal{R}_\phi(z) \subseteq A$ denote the set of rationalizable actions at z

Lemma 1

h satisfies plausible deniability if and only if for all $(x, y) \in \text{supp}(h)$,

$$\mathcal{R}_h(x, y) \subseteq \mathcal{R}_f(x).$$

Lemma 2

Given $\phi : \Omega \rightarrow \Delta(Z)$ and $z \in \text{supp}(\phi)$,

- (i) If $\phi(z|\omega)$ is constant in ω , $\mathcal{R}_\phi(z) = \{a_0\}$.
- (ii) If $\phi(z|\omega)$ is decreasing and non-constant in ω , $\mathcal{R}_\phi(z) = \{a : a \leq a_0\}$.
- (iii) If $\phi(z|\omega)$ is increasing and non-constant in ω , $\mathcal{R}_\phi(z) = \{a : a \geq a_0\}$.
- (iv) If $\phi(z|\omega)$ is non-monotone in ω , $\mathcal{R}_\phi(z) = A$.

Lemma 2

Given $\phi : \Omega \rightarrow \Delta(Z)$ and $z \in \text{supp}(\phi)$,

- (i) If $\phi(z|\omega)$ is constant in ω , $\mathcal{R}_\phi(z) = \{a_0\}$.
- (ii) If $\phi(z|\omega)$ is decreasing and non-constant in ω , $\mathcal{R}_\phi(z) = \{a : a \leq a_0\}$.
- (iii) If $\phi(z|\omega)$ is increasing and non-constant in ω , $\mathcal{R}_\phi(z) = \{a : a \geq a_0\}$.
- (iv) If $\phi(z|\omega)$ is non-monotone in ω , $\mathcal{R}_\phi(z) = A$.

Lemma 2

Given $\phi : \Omega \rightarrow \Delta(Z)$ and $z \in \text{supp}(\phi)$,

- (i) If $\phi(z|\omega)$ is constant in ω , $\mathcal{R}_\phi(z) = \{a_0\}$.
- (ii) If $\phi(z|\omega)$ is decreasing and non-constant in ω , $\mathcal{R}_\phi(z) = \{a : a \leq a_0\}$.
- (iii) If $\phi(z|\omega)$ is increasing and non-constant in ω , $\mathcal{R}_\phi(z) = \{a : a \geq a_0\}$.
- (iv) If $\phi(z|\omega)$ is non-monotone in ω , $\mathcal{R}_\phi(z) = A$.

Partitioning Message Space

- Partition X according to how $f(x|\omega)$ varies in ω :
 - (i) **Decreasing message**: $d \in D \subseteq X$, for which $f(d|\omega)$ is decreasing in ω
 - (ii) **Increasing message**: $i \in I \subseteq X$, for which $f(i|\omega)$ is increasing in ω
 - (iii) **Non-monotone message**: $s \in S \subseteq X$, for which $f(s|\omega)$ is non-monotone in ω

Theorem 1

h satisfies PD iff for all $y \in Y$,

- for each $d \in D$, $h(d, y|\omega)$ is decreasing in ω ,
 - for each $i \in I$, $h(i, y|\omega)$ is increasing in ω ,
 - for each $s \in S$, $h(s, y|\omega)$ is unrestricted.
-
- $f : \Omega \rightarrow \Delta(X)$ is **directional** if $S = \emptyset$; and **almost-directional** if $|S| \leq 1$.
 - We focus on the case where f is almost-directional.

Partitioning Message Space

- Partition X according to how $f(x|\omega)$ varies in ω :
 - (i) **Decreasing message**: $d \in D \subseteq X$, for which $f(d|\omega)$ is decreasing in ω
 - (ii) **Increasing message**: $i \in I \subseteq X$, for which $f(i|\omega)$ is increasing in ω
 - (iii) **Non-monotone message**: $s \in S \subseteq X$, for which $f(s|\omega)$ is non-monotone in ω

Theorem 1

h satisfies PD iff for all $y \in Y$,

- for each $d \in D$, $h(d, y|\omega)$ is decreasing in ω ,
- for each $i \in I$, $h(i, y|\omega)$ is increasing in ω ,
- for each $s \in S$, $h(s, y|\omega)$ is unrestricted.

- $f : \Omega \rightarrow \Delta(X)$ is **directional** if $S = \emptyset$; and **almost-directional** if $|S| \leq 1$.
 - We focus on the case where f is almost-directional.

Partitioning Message Space

- Partition X according to how $f(x|\omega)$ varies in ω :
 - (i) **Decreasing message**: $d \in D \subseteq X$, for which $f(d|\omega)$ is decreasing in ω
 - (ii) **Increasing message**: $i \in I \subseteq X$, for which $f(i|\omega)$ is increasing in ω
 - (iii) **Non-monotone message**: $s \in S \subseteq X$, for which $f(s|\omega)$ is non-monotone in ω

Theorem 1

h satisfies PD iff for all $y \in Y$,

- for each $d \in D$, $h(d, y|\omega)$ is decreasing in ω ,
- for each $i \in I$, $h(i, y|\omega)$ is increasing in ω ,
- for each $s \in S$, $h(s, y|\omega)$ is unrestricted.

- $f : \Omega \rightarrow \Delta(X)$ is **directional** if $S = \emptyset$; and **almost-directional** if $|S| \leq 1$.
 - We focus on the case where f is almost-directional.

Partitioning Message Space

- **Example:** $\Omega = \{\omega_1, \omega_2, \omega_3\}$ and $X = D \cup S \cup I$

	d_1	d_2	s	i_1	i_2
ω_1	0.4	0.2	0.2	0.1	0.1
ω_2	0.2	0.1	0.3	0.2	0.2
ω_3	0	0.1	0.2	0.4	0.3

$d_1, d_2 \in D$

$s \in S$

$i_1, i_2 \in I$

Direction-ordered Signal Representation

- A signal representation $\psi : \Omega \times [0, 1] \rightarrow X$ of f is **direction-ordered** if there exists functions $T_1, T_2 : \Omega \rightarrow [0, 1]$ with $T_1(\omega) \leq T_2(\omega)$ for all $\omega \in \Omega$ such that

$$\psi(\omega, t) \in \begin{cases} D, & \text{if } t < T_1(\omega), \\ S, & \text{if } T_1(\omega) \leq t < T_2(\omega), \\ I, & \text{if } T_2(\omega) \leq t. \end{cases}$$

Theorem

Given an almost-directional f , there exists a direction-ordered signal representing the SPD-greatest element.

Direction-ordered Signal Representation

- A signal representation $\psi : \Omega \times [0, 1] \rightarrow X$ of f is **direction-ordered** if there exists functions $T_1, T_2 : \Omega \rightarrow [0, 1]$ with $T_1(\omega) \leq T_2(\omega)$ for all $\omega \in \Omega$ such that

$$\psi(\omega, t) \in \begin{cases} D, & \text{if } t < T_1(\omega), \\ S, & \text{if } T_1(\omega) \leq t < T_2(\omega), \\ I, & \text{if } T_2(\omega) \leq t. \end{cases}$$

Theorem

Given an almost-directional f , there exists a direction-ordered signal representing the SPD-greatest element.

Example: SPD-greatest information

	y_1	y_2	y_3	y_4	y_5	y_6	y_7	y_8
ω_1	d_2	d_1	d_2	d_1	s	s	i_2	i_1
ω_2	d_2	d_1	s	s	i_2	i_1	i_2	i_1
ω_3	d_2	s	i_2	i_1	i_2	i_1	i_2	i_1
	.1	.2	.1	.2	.1	.1	.1	.1
	$T_1(\omega_3)$	$T_1(\omega_2) = T_2(\omega_3)$		$T_1(\omega_1) = T_2(\omega_2)$		$T_2(\omega_1)$		

Observations

1. **Binary State**: any baseline information structure is directional.
2. **Binary Message Space**: if the baseline information structure f satisfies the MLRP, then f is directional.

Corollary

In both cases, the SPD-greatest element can be represented by a direction-ordered signal.

Observations

1. **Binary State:** any baseline information structure is directional.
2. **Binary Message Space:** if the baseline information structure f satisfies the MLRP, then f is directional.

Corollary

In both cases, the SPD-greatest element can be represented by a direction-ordered signal.

Observations

1. **Binary State**: any baseline information structure is directional.
2. **Binary Message Space**: if the baseline information structure f satisfies the MLRP, then f is directional.

Corollary

In both cases, the SPD-greatest element can be represented by a direction-ordered signal.

Literature & Conclusion

- **Perfect Secrecy & Information Representation:**

- Shannon (1949): Foundational definition of perfect secrecy.
- Green & Stokey (1978): Signal representations on the unit interval.
- Cavounidis et al. (2025): Full revelation under perfect secrecy.

- **Information Design with Privacy/Independence Constraints:**

- Strack (2024), He et al. (2026): Blackwell frontiers under privacy and private-private information constraints.

- **Interacting Information Sources:**

- Börgers et al. (2013), Cheng, Börgers (2024): Complementarity and diversity of information sources.

- **Summary**
 - **Plausible Deniability (PD):** Communication can only nudge receivers in directions supported by baseline data.
 - **Optimal Design:** A simple **direction-ordered** signal achieves the SPD-greatest communication for almost-directional baselines.
- **In the paper:**
 - General Characterization of PD-greatest information
 - A sufficient condition that PD-greatest information is also SPD-greatest information

- **Summary**
 - **Plausible Deniability (PD):** Communication can only nudge receivers in directions supported by baseline data.
 - **Optimal Design:** A simple **direction-ordered** signal achieves the SPD-greatest communication for almost-directional baselines.
- **In the paper:**
 - General Characterization of PD-greatest information
 - A sufficient condition that PD-greatest information is also SPD-greatest information

Appendix

2 by 2 Example: Optimal Secret Information (w/o PD)

- Without Plausible Deniability, the optimal secret information is implemented by an *unordered* signal

